

QUESTION 1:

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// QUESTION 1 //
#include <iostream>
using namespace std;
struct student{
    string name;
    int ID;
    int sub_count;
    float scores[5];
};
void processPersonData(student s1){
    float sum=0;
    float avg;
    int second_largest;
    cout<<"====DISPLAY RESULTS===="<<endl;
    cout<<"NAME: "<<s1.name<<endl;
    cout<<"ID: "<<s1.ID<<endl;
    cout<<"GRADES: ";
    for(int i=0;i<s1.sub_count;i++){
        cout<<s1.scores[i];
        cout<<" ";
        sum+=s1.scores[i];
    }
    cout<<endl;
    avg=sum/s1.sub_count;
    cout<<"AVERAGE: "<<avg;
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    avg=sum/s1.sub_count;
    cout<<"AVERAGE: "<<avg;
    cout<<endl;
    float largest=s1.scores[0];
    for(int j=0;j<s1.sub_count;j++){
        if(s1.scores[j]>largest){
            largest=s1.scores[j];
        }
    }
    float sec_largest=s1.scores[0];
    for(int k=0;k<s1.sub_count;k++){
        if(s1.scores[k]>sec_largest && s1.scores[k]<largest){
            sec_largest=s1.scores[k];
        }
    }

    cout<<"second largest: "<<sec_largest;
}

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int main(){
    student s1;
    student *s1_ptr=&s1;
    cout<<"ENTER STUDENT NAME:"<<endl;
    cin>>s1_ptr->name;
    cout<<"ENTER STUDENT ID:"<<endl;
    cin>>s1_ptr->ID;
    cout<<"ENTER NUMBER OF SCORES:"<<endl;
    cin>>s1_ptr->sub_count;
    cout<<"ENTER SCORE:"<<endl;
    for(int i=0;i<s1_ptr->sub_count;i++){
        cout<<"SUB "<<i+1<<endl;
        cin>>s1_ptr->scores[i];
    }
    processPersonData(s1);
}

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C:\Sem 2\001-ENBS7C4>C:\vs
ENTER STUDENT NAME:
waleeja
ENTER STUDENT ID:
0005
ENTER NUMBER OF SCORES:
3
ENTER SCORE:
SUB 1
23
SUB 2
45
SUB 3
76
```

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=====DISPLAY RESULTS=====
NAME: waleeja
ID: 5
GRADES: 23 45 76
AVERAGE: 48
second largest: 45
C:\Sem 2\001-ENBS7C4>
```

QUESTION 2:

```
#include <iostream>
#include <string>
using namespace std;
struct Emp{
    public:
        char employeeName[50];
        char jobTitle[30];
        int employeeCode;
        int monthCount;
        float monthlySalary[5];
        float *base_address;
};
void calculate_display_Payroll(Emp *ptr_to_e,float *monthlySalary,int extra){
    float sum=0;
    float highest=monthlySalary[0];

    for(int i=0;i<ptr_to_e->monthCount;i++){
        sum+=monthlySalary[i];
    }
    float avg=sum/ptr_to_e->monthCount;

    for(int i=0;i<ptr_to_e->monthCount;i++){
        if(monthlySalary[i]>highest){
            highest=monthlySalary[i];
        }
    }
}
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    for(int i=0;i<ptr_to_e->monthCount;i++){
        if(monthlySalary[i]>highest){
            highest=monthlySalary[i];
        }
    }

    //DISPLAY
    cout << "--- PAYROLL REPORT ---"<<endl;

    cout<<"EMPLOYEE NAME: "<<ptr_to_e->employeeName<<endl;
    cout<<"JOB TITLE: "<<ptr_to_e->jobTitle<<endl;
    cout<<"EMPLOYEE CODE: "<<ptr_to_e->employeeCode<<endl;
    cout<<"MONTHLY SALARY: "<<endl;
    for(int i=0;i<ptr_to_e->monthCount;i++){
        cout<<"MONTH "<<i+1<<" ";
        cout<<ptr_to_e->monthlySalary[i];
        cout<<endl;
    }

    cout<<"AVERAGE SALARY : "<<avg<<endl;
    cout<<"HIGHEST MONTHLY SALARY: "<<highest<<endl;
    cout<<"AVERAGE SALARAY + EXTRA CHARGES: "<<avg+extra<<endl;

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}

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int main(){
    Emp e;
    Emp *ptr_to_e=&e;
    int extra_amount;
    cout<<"EMPLOYEE NAME: ";
    cin.getline(ptr_to_e->employeeName, sizeof(ptr_to_e->employeeName));

    cout<<"JOB TITLE: ";
    cin.getline(ptr_to_e->jobTitle, sizeof(ptr_to_e->jobTitle));

    cout<<"EMPLOYEE CODE: ";
    cin>>ptr_to_e->employeeCode;

    cout<<"MONTH COUNT: ";
    cin>>ptr_to_e->monthCount;

    cout<<"MONTHLY SALARY: "<<endl;
    for(int i=0;i<ptr_to_e->monthCount;i++){
        cout<<"MONTH "<<i+1<<" ";
        cin>>ptr_to_e->monthlySalary[i];
        cout<<endl;
    }
}

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    }
    cout<<"ENTER EXTRA AMOUNT: ";
    cin>>extra_amount;
    ptr_to_e->base_address=ptr_to_e->monthlySalary;

    calculate_display_Payroll(ptr_to_e,ptr_to_e->base_address,extra_amount);

}

```

EMPLOYEE NAME: waleeja

JOB TITLE: hr manager

EMPLOYEE CODE: 345

MONTH COUNT: 3

MONTHLY SALARY:

MONTH 1 23.56

MONTH 2 4876.90

MONTH 3 3456.8

ENTER EXTRA AMOUNT: 670

--- PAYROLL REPORT ---

EMPLOYEE NAME: waleeja

JOB TITLE: hr manager

EMPLOYEE CODE: 345

MONTHLY SALARY:

MONTH 1 23.56

MONTH 2 4876.9

MONTH 3 3456.8

AVERAGE SALARY : 2785.75

HIGHEST MONTHLY SALARY: 4876.9

AVERAGE SALARAY + EXTRA CHARGES: 3455.75